

LayerIt! — Commented Thesis Outline

Merged 6-chapter hypothesis structure

Proposed outline (merged, 6 chapters)

1 Introduction

- 1.1 Contextualization
- 1.2 Motivation
- 1.3 Objectives, Research Questions and Contributions
- 1.4 Developed Project
- 1.5 Thesis Structure

2 Literature Review

- 2.1 Visualization Types (2.1.1 Sheet Music · 2.1.2 Symbolic · 2.1.3 Audio)
- 2.2 MIR Visualization Tools
- 2.3 Audio-to-Score Alignment
- 2.4 Summary

3 Methodology

- 3.1 Research Approach
- 3.2 Iterative Development Process (3.2.1–3.2.5, the five prototypes)
- 3.3 Scope Decision
- 3.4 Evaluation Criteria

4 LayerIt: Design and Implementation

- 4.1 The Problem and our Solution
- 4.2 LayerIt Overview
- 4.3 Design Choices and Rationale
- 4.4 Generating Visuals (4.4.1 Warped Score · 4.4.2 MAPS · 4.4.3 BT data)
- 4.5 Layer Alignment Method
- 4.6 Layers on Layers, within Layers (4.6.1 Visual · 4.6.2 SVG · 4.6.3 Code)
- 4.7 Chapter Summary

5 Evaluation

- 5.1 Evaluation Approach
- 5.2 Reproducing Composite Figure Types (RQ1)
- 5.3 Workflow Comparison: Annotation and Evaluation (RQ2, RQ3)
- 5.4 Extensibility Probe (RQ2)
- 5.5 Limitations and Difficult Cases (RQ2, RQ3)
- 5.6 Chapter Summary

6 Conclusions and Future Work

- 6.1 Achievements
- 6.2 Future Work
- 6.3 Final Remarks

Alternative choices

- **Name Chapter 5 “Validation” instead of “Evaluation.”** Defensible given there is no user study and the chapter mostly demonstrates the tool against stated criteria. “Evaluation” is broader and fits the comparison tables and effort metrics; “Validation” signals more modest claims. Either works — pick one and use it consistently in the RQ forward-references.
- **Keep the 7-chapter split instead of merging.** Split Chapter 4 back into “Developed Solution” (4.1–4.3) and “Implementation” (4.4–4.6). Choose this if Chapter 4 grows too long once written; the natural break is the 4.3/4.4 boundary.
- **Methodology as a section, not a chapter.** Fold Chapter 3 into the start of Chapter 4 as §4.1 “Methodology.” Lighter-touch (yields 5 chapters); weaker signal to an examiner. Only if the department template discourages short chapters.

- **RQs last in the intro.** Order 1.1 Context → 1.2 Motivation → 1.3 Developed Project → 1.4 Objectives/RQs → 1.5 Structure, if the RQs work better as the sharp closing statement the introduction builds toward.
- **Merge 5.4 into 5.2/5.3.** The extensibility probe (now an RQ2 dimension) could be a subsection rather than its own section if the result is short.
- ...

Comments (Goal + content to keep / content to correct)

Content-level only. All English, typo, naming, and mechanical fixes live in the separate corrections document.

Chapter 1 — Introduction

1.1 Contextualization — *Goal:* establish MIR and why visualization matters.

- **Keep:** the MIR-as-multidisciplinary framing and the “how/what we visualize shapes conclusions” thesis.
- **Correct:** scope it to field + general problem; move thesis-specific drivers to Motivation so the two don't overlap once adjacent.

1.2 Motivation (moved up) — *Goal:* make the case for the work before any solution is described.

- **Keep:** the BT/BA annotation-and-evaluation pain points and the SV-limitations argument — the strongest part of the current intro. This is also where the annotation-infrastructure origin story belongs (told honestly, as motivation).
- **Correct:** now that it precedes the objectives, end it by pointing at the gap the RQs will target, not at the solution.

1.3 Objectives, Research Questions and Contributions (new) — *Goal:* state the objectives (O1–O3), the research questions (RQ1–RQ3), and an explicit contributions list.

- **Content:** use the O1–O3 / RQ1–RQ3 block from the drafts file. Objectives are the build-and-demonstrate spine; RQs are phrased as “how / to what extent” so answers are graded, not yes/no. Frame the contribution as the OO layer-composition model plus the figures it enables, explicitly not the alignment (which is ScoreWarp's).
- **Content:** limits are folded into RQ2 and RQ3 rather than a separate RQ. Extensibility is an evaluation dimension under RQ2, not its own question.
- **Content:** contributions list (tie to O1/O3, restated in §6.1): the LayerIt framework; its release as an open-source, pip-installable library; the open-source repository of the five prototypes; a Late-Breaking/Demo paper submitted to ISMIR 2026 (under review — phrase precisely, never “accepted”).
- **Content:** annotation appears here only as an assessed workflow (RQ3/O3), never as a delivered system.

1.4 Developed Project (moved down) — *Goal:* high-level sketch of LayerIt as solution.

- **Keep:** the concise “what we built” description and the layer/SVG summary.
- **Correct:** make it read as a preview following the objectives/RQs, not the first thing after context.

1.5 Thesis Structure — *Goal:* roadmap.

- **Correct:** reflect the new Methodology and Evaluation chapters and the merged Ch. 4.

Chapter 2 — Literature Review

2.1 Visualization Types — *Goal:* categorize sheet-music / symbolic / audio representations.

- **Keep:** the three-way taxonomy, the MEI/Verovio explanation, and all the audio-representation figures (waveform → spectrogram variants → CQT → chromagram). Solid, well-illustrated groundwork.
- **Correct:** the chromagram pitch-class count — 12, not 11 (a factual error; also in corrections).

2.2 MIR Visualization Tools — *Goal:* contrast GUI tools with code workflows; set up the gap.

- **Keep:** the SV-vs-code argument, the reproducibility/ML rationale for code-based workflows, and the PP framing — this directly motivates the thesis.

2.3 Audio-to-Score Alignment — *Goal:* define alignment and establish the terminology used downstream.

- **Keep:** offline/online distinction, note/beat/segment levels, ScoreWarp introduction.
- **Correct:** replace the improvised “strict/loose” terms with a literature-grounded pair — time-warped (continuous) vs. link-based (anchor) alignment. Introduce both here; every later chapter depends on it.

2.4 Summary — *Goal:* close and bridge to methodology.

- **Keep:** as-is.

Chapter 3 — Methodology (new)

3.1 Research Approach — *Goal*: name the method (exploratory, iterative research-through-design).

- **Content**: new, short. Frame the artifact as both outcome and instrument of inquiry — the direction changed because building revealed the real gap, which is a legitimate method, not a wobble.

3.2 Iterative Development Process — *Goal*: present the five prototypes as design cycles.

- **Keep**: all factual content of the current iteration write-ups (tech used, what was built, why dropped) — already there and genuinely interesting.
- **Correct**: reframe each as goal → built → lesson → decision carried forward, so knowledge visibly accumulates toward LayerIt. Add the summary table.

3.3 Scope Decision — *Goal*: record the deliberate narrowing from “BA/BT GUI” to “OO visualization library.”

- **Content**: consolidate the scope-shift reasoning currently scattered across §3.2. Key claim: interaction and visualization couldn't both be solved in one thesis, and visualization was the under-served, higher-value gap.

3.4 Evaluation Criteria — *Goal*: state the criteria Ch. 5 will apply.

- **Content**: new. Coverage, manual-effort reduction, reproducibility, extensibility, fidelity/limits. Defining them here is what makes the evaluation read as deliberate rather than a demo.

Chapter 4 — LayerIt: Design and Implementation (merged)

4.1 The Problem and our Solution — *Goal*: restate the concrete problem and LayerIt's answer.

- **Keep**: the two-modes-of-ad-hoc-visualization framing (code-based vs. manual image editing).
- **Correct**: trim overlap with §3.3 — keep this about the solution. Per the supervisor's framing, be honest that the alignment layer is largely a ScoreWarp encapsulation; the thesis-worthy contribution is the OO approach plus enabled figures.

4.2 LayerIt Overview — *Goal*: what LayerIt is and the libraries it builds on.

- **Keep**: as-is.

4.3 Design Choices and Rationale — *Goal*: justify SVG-first, library choices, SVG/PNG trade-off.

- **Keep**: the vector-vs-raster reasoning and the “SVG as language-agnostic bridge” argument — a strong justification section.
- **Correct**: relocate here so the SVG rationale is stated once (currently half-repeats between old §3.3.2 and Ch. 4).

4.4 Generating Visuals — *Goal*: required input files and how to get them.

- **Keep**: Table 4.1, the MAPS structure listing and field explanations, the BT .npz field table — clear, useful reference content.

4.5 Layer Alignment Method — *Goal*: the pixel/time rule-of-three against the warped-score timeline.

- **Keep**: as-is; the method is well explained.

4.6 Layers on Layers, within Layers — *Goal*: the “layers within layers” philosophy at visual / SVG / code levels.

- **Keep**: the three-level articulation, Figures 4.1 and 4.3, and the code listings — the conceptual core of the contribution.
- **Correct**: move current Fig. 4.2 (BeatThis output vs. probability curve) out of here into Evaluation §5.3, where it becomes the evaluation-workflow centerpiece. Its “side note” analysis is really an evaluation result, not an implementation detail.

4.7 Chapter Summary — *Goal*: close design and implementation.

- **Correct**: merge the two old chapter summaries (§3.4 + §4.4) into one to avoid repetition.

Chapter 5 — Evaluation (new)

5.1 Evaluation Approach — *Goal*: explain the three means plus the “from our own data” rule.

- **Content**: new. State plainly there's no user study; evaluation is by demonstration, comparison, and an extensibility probe against the §3.4 criteria.
- **Content**: figures are generated from own data; published figures are referenced only as motivation. The unpublished (Rev98) figure must not be reproduced — build that figure type from own data.

5.2 Reproducing Composite Figure Types (RQ1) — *Goal*: show LayerIt reproduces literature figure types in code.

- **Content**: four sub-cases — score+chroma; the Sá Pinto “gap” case (two figures collapsed into one); probability+positions stack; waveform+positions. The Sá Pinto case is the centerpiece (their figure needs a separate score figure for bar context — LayerIt removes that need).
- **Needs**: the four figures generated from own data; the coverage table.

5.3 Workflow Comparison (RQ2 + RQ3) — *Goal*: compare LayerIt to SV, PP, manual SVG editing.

- **Keep:** promote current Fig. 4.2 here as the evaluation-workflow example.
- **Content:** the annotation-workflow argument hinges on time-warped vs. link-based — for annotators, the score must sit physically under the time axis (PP's connecting lines don't). Feature table plus honest effort/time table (separate one-time setup from per-figure cost; report a manual-operation count alongside wall-clock). The annotation strand answers RQ3; the effort/simplification strand answers RQ2.

5.4 Extensibility Probe (RQ2) — *Goal:* measure the real cost of adding one layer.

- **Needs:** actually implement a new layer; report line count; confirm zero core changes. “Added layer X in N lines without touching the core” is the checkable result. This answers the “how readily can new visualizations be added?” clause of RQ2.

5.5 Limitations and Difficult Cases (RQ2 + RQ3) — *Goal:* document where the approach degrades.

- **Needs:** run an expressive/rubato piece and show the breakdown; show sparse-vs-dense onset interpolation error.
- **Content:** answers the “under what conditions does that support break down?” clause of RQ3 and the ceiling of RQ2's simplification claim. The final-SVG-hand-editing note belongs here as a fallback, not a feature — it sacrifices reproducibility and reintroduces the manual process LayerIt replaces.

5.6 Chapter Summary — *Goal:* summarize RQ1–RQ3 answers.

Chapter 6 — Conclusions and Future Work

6.1 Achievements — *Goal:* close on the objectives (did we achieve O1–O3?), each citing its RQ's evidence, then the contributions.

- **Keep:** the seven current achievements supply the raw material.
- **Correct:** restructure so the answer is by objective — O1 (framework built), O2 (difficult-with-current-tools workflows demonstrated), O3 (usefulness for annotation/evaluation assessed) — each cross-referencing the RQ evidence in Ch. 5. The open-source repos, pip package, and ISMIR demo resurface here as proof O1/O3 were met, not just asserted. Use the rewrite in the drafts file.

6.2 Future Work — *Goal:* evolutionary directions.

- **Keep:** the current seven directions are strong (interactive frontends, expanded coverage, ecosystem integration, collaborative annotation, comparative algorithm viz, scalability, standardization). This is where the rest of the original annotation-infrastructure vision legitimately lives. The Morais “Skip That Beat” retraining example fits the comparative/coverage item.

6.3 Final Remarks — *Goal:* broader significance.

- **Keep:** as-is.